m = 4 Macaulay matrix: fitted w and the repriced degree-5 row

Status. Fitted w from 1561 recorded closure rows, plus 34 new matrix-profile runs (m = 4, five cells, degrees 3 and 4). The profiler's word ops equal the recorded ones on 29 of 29 comparable rows. S3 gate run by the user (session network policy refused it). No new solver, no speed claim; rho stays 2^60.81.

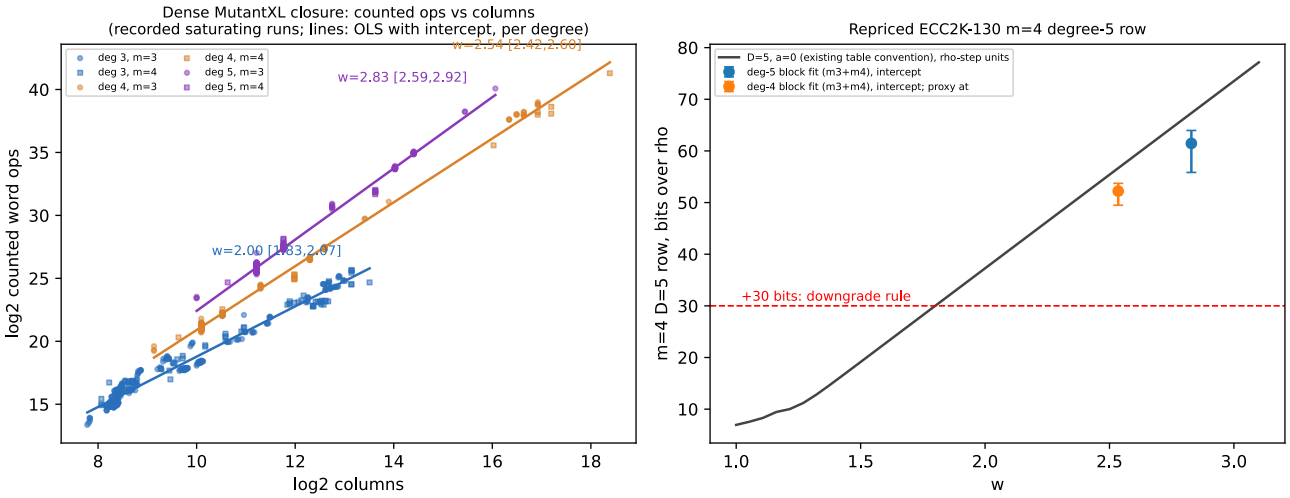


Figure 1: Left: counted word ops vs columns, saturating closures, OLS with intercept per degree. Right: D=5 row in rho-step units vs w; points are the fitted (a, w) with bootstrap 95% intervals.

Fitted w (cell-clustered bootstrap, intercept included)

block	rows / cells	log2 cols	w	95% CI
degree 3, m=3+4	495 / 37	7.8 to 13.5	2.00	1.83 to 2.07
degree 4, m=3+4	367 / 18	9.1 to 18.4	2.54	2.42 to 2.60
degree 5, m=3+4	345 / 12	10.0 to 16.1	2.83	2.59 to 2.92
pooled (degree intercepts)	1207 / 37		2.34	2.16 to 2.40

The local slope at the top of the range is 2.7 (degree 4) and 3.2 (degree 5): w rises toward the dense cubic regime. Degree 3 and degree 4 do not overlap.

D = 5 row, m = 4, bits over rho

price	A: column touches	B: rho steps
w = 1 floor (existing +10.32)	+10.32	+6.95
w = 1.80 (crosses +30)	+38.45	+30.02
w = 2.00, a = 0	+45.71	+37.27
degree-4 proxy (w 2.54)	+60.65	+52.21 (49.5 to 53.7)
degree-5 fit (w 2.83)	+69.89	+61.45 (55.8 to 64.0)

Matrix profile (degree 4, measured)

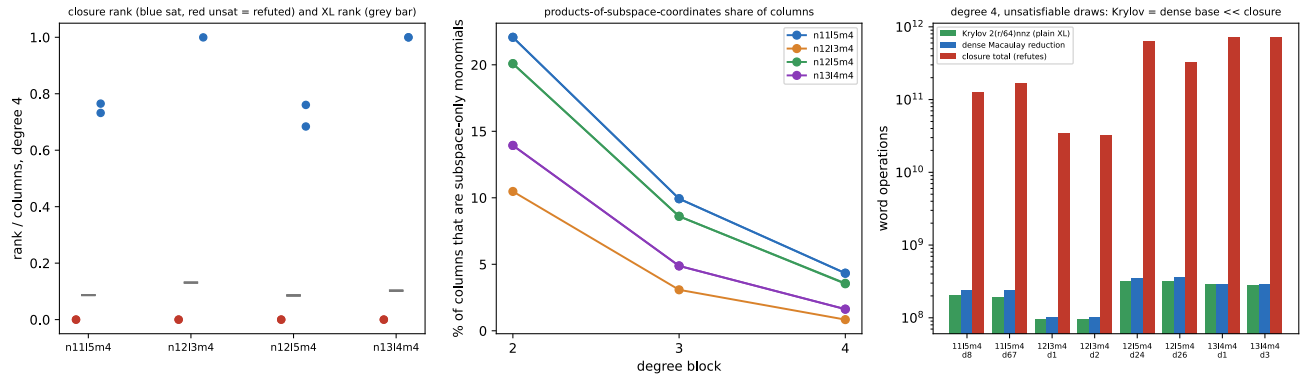


Figure 2: Left: closure rank over columns (satisfiable 0.68 to 1.00; unsatisfiable collapse to rank 1) against plain XL rank (0.09 to 0.13). Middle: share of columns that are subspace-only monomials, by degree block. Right: Krylov cost of the plain XL matrix equals the dense Macaulay reduction and is 340 to 2500 times below the closure, which is what refutes.

cell (n,l,N)	cols	closure rank / cols	XL rank / cols	rho	closure vs base ops
(11,5,42)	124314	0.73 to 0.77	0.087	52 to 56	3.2e11 vs 2.4e8
(12,5,44)	149905	0.68 to 0.76	0.086	59 to 61	2.9e11 vs 3.6e8
(13,4,42)	124314	1.000	0.102	54 to 56	6.1e11 vs 2.9e8
(12,3,36)	66712	1.000	0.131	40	5.1e10 vs 1.0e8

Subspace-only columns are 22% of degree-2 columns and 1.6% to 4.3% of degree-4 columns: almost every column touches an intermediate-point coordinate. The closure is over 99.7% of the work.

Structured elimination, priced at the real row (n = 131, N = 398)

plain-XL degree D’	Krylov word ops	row, bits over rho (conv −8.4)	no conversion
6 (measured one-degree lag)	2^78.0 to 2^78.7	+42.8 to +43.5	+51.2 to +51.9
5 (needs XL = closure degree; 0 of 136 support it)	2^64.8 to 2^65.5	+29.6 to +30.3	+38.0 to +38.7

At the measured sizes the Krylov cost equals the dense Macaulay reduction (ratio 0.8 to 1.0); a structured solve lowers the row from +61.5 to about +43 through a constant, not an exponent below 2.

Decision

Above +30 under every measured pricing, including the structured route (about +43): the (4,6) cell is a degree curiosity; do not rent memory for it. The Krylov route reaches the +30 line only if plain XL refuted at the closure’s own degree, which 0 of 136 committed draws support.